* **Regla de SARRUS** (matrices 3 x 3) $\quad \det = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = a \cdot d - b \cdot c$

Ejerc. $\quad M = \begin{pmatrix} 2 & 1 & 3 \\ -1 & 4 & 0 \\ 6 & -2 & 1 \end{pmatrix}$

$$\det(M) = \begin{vmatrix} 2 & 1 & 3 \\ -1 & 4 & 0 \\ 6 & -2 & 1 \\ 2 & 1 & 3 \\ -1 & 4 & 0 \end{vmatrix} = 2.4.1 + (-1).(-2).3 + 6.1.0 - [6.4.3 + 2.(-2).0 + (-1).1.1]$$

$$= 8 + 6 - [72 - 1] = 14 - 71 = \boxed{-57}$$

* **Mét. de los ADJUNTOS o COFACTORES** (para orden $n \geq 3$)

Ejerc. $\quad A = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 3 & -1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 4 & 0 & -2 & 1 \end{pmatrix}$

$$\det A = a_{14} \cdot C_{14} + a_{24} \cdot C_{24} + a_{34} \cdot C_{34} + a_{44} \cdot C_{44}$$

$$C_{ij} = (-1)^{i+j} \cdot \text{Menor}_{ij}$$

$$\det(A) = \underbrace{\quad\quad}_{} 0 \underbrace{\quad\quad}_{} + 1 \cdot (-1)^{3+4} \cdot \begin{vmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \\ 4 & 0 & -2 \end{vmatrix} + 1 \cdot (-1)^{4+4} \cdot \begin{vmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \\ 0 & 2 & 0 \end{vmatrix}$$

$$= -1 \cdot [1.(-1).(-2) - (-8)] + 12 - 2$$

$$-1 \cdot 10 + 10$$

$$-10 + 10 = \boxed{0}$$

Ejerc. Se nos da $A = (A_1, A_2, A_3) \in \mathbb{R}^{3\times 3}$ con $\det(A) = -3$

Calcular $\det\left(\frac{3}{4} \cdot A^{-1} \cdot B^T\right)$ donde $B = (A_1 - 3A_3, A_3, A_1 - A_2)$.

- Prop. de determ.

$\det\left(\frac{3}{4} \cdot A^{-1} \cdot B^T\right) = \left(\frac{3}{4}\right)^3 \cdot \det(A^{-1} \cdot B^T)$ Prop. $\det(kM) = k^n \cdot \det(M)$

$= \frac{27}{64} \cdot \det(A^{-1}) \cdot \det(B^T)$. Prop. $\det(MN) = \det M \cdot \det N$

$= \frac{27}{64} \cdot \frac{1}{\det(A)} \cdot \det(B)$ Prop.

$\det(B) = \det(A_1 - 3A_3, A_3, A_1 - A_2)$ $C_1 \to C_1 + 3C_2$

$\quad = \det(A_1, A_3, A_1 - A_2)$ $C_3 \to C_3 - C_1$

$\quad = \det(A_1, A_3, -A_2)$ Factorizar escalar de una columna.

$\quad = (-1)\cdot(-1) \cdot \det(A_1, A_2, A_3)$ Intercambiar $C_2 \leftrightarrow C_3$

$\det(B) = \det(A) = -3$

$$= \frac{27}{64} \cdot \frac{1}{\cancel{-3}} \cdot (\cancel{-3}) = \boxed{\frac{27}{64}}$$

Ejerc. Sea $A \in \mathbb{R}^{4\times 4}$ con $\det(A) = k \neq 0$. Calcule $\det\left(\frac{2}{3} \cdot B^{-1} \cdot A^2\right)$

Si $B = (A_1 - A_2, A_3 - 2A_4, A_3, A_2)$.

$C_1 \to C_1 + C_4$: $\det(B) = \det(A_1, A_3 - 2A_4, A_3, A_2)$

$C_2 \to C_2 - C_3$: $= \det(A_1, -2A_4, A_3, A_2)$

$C_4 \leftrightarrow C_2$ $= -2 \cdot \det(A_1, A_4, A_3, A_2)$

$= -2 \det(A)$

$= -2 \cdot k$

$$\det\left(\frac{2}{3}\cdot B^{-1}\cdot A^{2}\right) = \left(\frac{2}{3}\right)^{4}\cdot \det(B^{-1})\cdot \det(A^{2})$$

$$= \frac{16}{81}\cdot\frac{1}{\det(B)}\cdot\left(\det(A)\right)^{2}$$

$$= \frac{\overset{8}{\cancel{16}}}{81}\cdot\frac{1}{-\cancel{2k}}\cdot k^{\cancel{2}} = \boxed{-\frac{8k}{81}}$$

· Ejerc. Encontrar $k\in\mathbb{R}$ para que A sea SINGULAR $(\det(A)=0)$

$$A=\begin{pmatrix} k-1 & -2 & 3 \\ k+2 & -1 & 0 \\ 0 & 0 & 3-k \end{pmatrix}$$

$$\det(A) = (k-1)\cdot(-1)\cdot(3-k) - \left[(3-k)\cdot(-2)\cdot(k+2)\right]$$

$$= (3-k)\cdot\left[-k+1 \quad +2k+4\right]$$

$$= (3-k)\cdot(k+5) = 0$$

Será singular para $k=3 \wedge k=-5$

· Ejerc. Halle los valores de k que hacen inversible a P $(\det(P)\neq 0)$

$$P=\begin{pmatrix} 1 & 1 & 0 \\ -2k-1 & 2 & 0 \\ 0 & 3 & k \end{pmatrix}$$

$$\det(P) = 2k - k\cdot(-2k-1)$$

$$= 2k + 2k^{2} + k = 0$$

$$2k^{2} + 3k = 0$$

$$k\cdot(2k+3) = 0$$

$$k=0 \vee k=-\frac{3}{2}$$

$\therefore$ P es inversible si $k\neq 0 \wedge k\neq -\frac{3}{2}$

. Ejerc.

$$\begin{cases} x + y + z = a \\ x - y = 0 \\ 3x + y + bz = 0 \end{cases} \Rightarrow A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 3 & 1 & b \end{pmatrix}$$

$$\det(A) = -b + 1 - [-3 + b] = \underline{-2b + 4}$$

. SCD $\Rightarrow -2b + 4 \neq 0$
$\left.\begin{matrix} \\ b \neq 2 \end{matrix}\right\}$ $\begin{matrix} a \in \mathbb{R} \\ b \in \mathbb{R} - \{2\} \end{matrix}$

. Si $\boxed{b = 2} \Rightarrow \det(A) = 0$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 1 & -1 & 0 & 0 \\ 3 & 1 & 2 & 0 \end{array}\right) \begin{matrix} F_2 : F_2 - F_1 \\ F_3 : F_3 - 3F_1 \end{matrix} \left(\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & -2 & -1 & -a \\ 0 & -2 & -1 & -3a \end{array}\right) F_3 : F_3 - F_2$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & -2 & -1 & -a \\ 0 & 0 & 0 & -2a \end{array}\right)$$

. SI $\Rightarrow -2a \neq 0$
$a \neq 0$

. SCI $\Rightarrow -2a = 0$
$\underline{a = 0}$

Ejerc.

$$\begin{cases} -2x - 4y + 6z = 4 \\ -3x - 3y + 8z = 4 \\ 3y - z = k \end{cases} \Rightarrow A = \begin{pmatrix} -2 & -4 & 6 \\ -3 & -3 & 8 \\ 0 & 3 & -1 \end{pmatrix}$$

$$\det(A) = -6 - 54 - [-48 - 12] = -60 + 60 = \boxed{0}$$ es SINGULAR

. SCD $\Rightarrow$ NUNCA

. Analices SI, SCI :

$$\left(\begin{array}{ccc|c} -2 & -4 & 6 & 4 \\ -3 & -3 & 8 & 4 \\ 0 & 3 & -1 & k \end{array}\right) \xrightarrow{F_1/-2} \left(\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ -3 & -3 & 8 & 4 \\ 0 & 3 & -1 & k \end{array}\right) \xrightarrow{F_2 : F_2 + 3F_1}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 0 & 3 & -1 & -2 \\ 0 & 3 & -1 & k \end{array}\right) \xrightarrow{F_3 : F_3 - F_2} \left(\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 0 & 3 & -1 & -2 \\ 0 & 0 & 0 & k+2 \end{array}\right)$$

SCI $\Rightarrow$ $k+2=0$; SI $\Rightarrow$ $k+2 \neq 0$

$k=-2$; $k \neq -2$

$$\begin{cases} x + 2y - 3z = -2 \quad (1) \\ 3y - z = -2 \Rightarrow z = 3y+2 \end{cases}$$

$$x + 2y - 3.(3y+2) = -2$$

$$x + 2y - 9y - 6 = -2$$

$$x = 7y + 4$$

Si $y = t$

$$(x, y, z) = (7t+4 ; t ; 3t+2)$$

$$= (7t ; t ; 3t) + (4 ; 0 ; 2)$$

$$= t.(7 ; 1 ; 3) + (4 ; 0 ; 2)$$